12/16/25

$$A=\begin{bmatrix}\boxed{1} & 2 & 3\\ 2 & 4 & 6\\ 2 & 6 & 8\\ 2 & 8 & 10\end{bmatrix} \rightarrow \begin{bmatrix}\boxed{1} & 2 & 3\\ 0 & 0 & 0\\ 0 & 2 & 2\\ 0 & 4 & 4\end{bmatrix} \xrightarrow{\text{row exchange}} \underbrace{\begin{bmatrix}\boxed{1} & 2 & 3\\ 0 & \boxed{2} & 2\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}}_{U}$$

rank = 2

2 pivot columns (1 and 2)
1 free column (3)

Nullspace

$x=\begin{bmatrix}-1\\-1\\1\end{bmatrix}$ is in nullspace

To check for Nullspace, start by giving free variable a convenient value, let's say 1.

$x_3 = 1$

$x_1+2x_2+3x_3=0$
$2x_2+2x_3=0$

↳ So the whole nullspace is $x=c\begin{bmatrix}-1\\-1\\1\end{bmatrix}$

(c is important)
(whole nullspace needs c)
(basis of nullspace is vector $\begin{bmatrix}-1\\-1\\1\end{bmatrix}$)

Keep going to R

$$\begin{bmatrix}\boxed{1} & 2 & 3\\ 0 & \boxed{2} & 2\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix} \rightarrow \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix} = R$$

I

↳ take 1 row 2 from row 1
↳ divide row 2 by 2

$x=c\begin{bmatrix}-1\\-1\\1\end{bmatrix}=c\begin{bmatrix}-F\\I\end{bmatrix}$

N (Nullspace Matrix)

Lecture #8, Solving Ax=b; Row Reduced Form R 12/17/25

$$x_1 + 2x_2 + 2x_3 + 2x_4 = b_1$$
$$2x_1 + 4x_2 + 6x_3 + 8x_4 = b_2$$
$$3x_1 + 6x_2 + 8x_3 + 10x_4 = b_3$$

$$\left[\begin{array}{cccc|c} \boxed{1} & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} \boxed{1} & 2 & 2 & 2 & b_1 \\ 0 & 0 & \boxed{2} & 4 & b_2-2b_1 \\ 0 & 0 & 2 & 4 & b_3-3b_1 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} \boxed{1} & 2 & 2 & 2 & b_1 \\ 0 & 0 & \boxed{2} & 4 & b_2-2b_1 \\ 0 & 0 & 0 & 0 & b_3-b_2-b_1 \end{array}\right]$$

Augmented Matrix $= [A \ b]$ pivot cols 1 and 3

$$\boxed{0 = b_3 - b_2 - b_1}$$

This is expected for solvability because the eqns of $b_1 + b_2 = b_3$

Try $b = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2-2b_1 \\ 0 & 0 & 0 & 0 & b_3-b_2-b_1 \end{array}\right] \Rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 0 & 0 & 2 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

so $b = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$ is OK since it allows for a solution

Solvability: Condition on RHS (b)

- $Ax = b$ is solvable when b is in column space of A → defines solvability
- If a combination of the rows of A gives zero row, then the same combinations of entries of b must give 0. → how to test for solvability

To find complete solution to $Ax=b$

① $x_{particular}$: set all free variables to zero then solve $Ax=b$ for pivot variables

In this case, x_2 & $x_4 = 0$ so in original eqns : $x_1 + 2x_3 = 1$, $2x_3 = 3$ $\quad x_1 = -2$, $x_3 = 3/2$

therefore $x_p = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}$

+

② $x_{null\ space}$

$$x_{complete} = \underbrace{\begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}}_{x_p} + \underbrace{c_1\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2\begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}}_{x_n}$$

(all comb. of special solutions from null space)

Complete solution $x = x_p + x_n$

$$Ax_p = b$$
$$Ax_n = 0$$
$$A(x_p + x_n) = b$$

If we have 1 solution, we can add on anything in nullspace because null space = 0 RHS and we will still have b.

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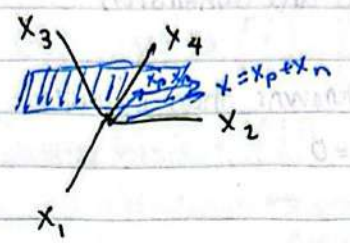
$r=m=n$	$r=n<m$	$r=m<n$	$r<m,\ r<n$
$R=I$	$R=\begin{bmatrix} I \\ 0 \end{bmatrix}$	$R=\begin{bmatrix} I & F \end{bmatrix}$	$R=\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
(1 Solution to $Ax=b$)	(0 or 1 solution)	(∞ Solutions)	(0 or ∞ solutions)

Plot all solutions x in R^4...



Drawing 4D world on 2D paper is impossible

BUT : just notice the planes

the bottom one through the origin → nullspace

top plane floating → solution set for $Ax=b$

vector x_p → started at origin & points to top plane

its job is to lift us out of null space and into solution plane

$x=x_p+x_n$ → shows that to find any solution on that top plane we do two steps, use elevator x_p to go up then walk around the flat surface x_n

Why does this work?

x_p solves b → elevates

x_n allows sideway movement on plane $(+0)$

Together $A(x_p+x_n)=b+0=b$

Think of m by n matrix of rank r

we always know that $r \le m$, $r \le n$

$r \le m$: m rows, so we can't have more than m pivots

$r \le n$: because the column can't have more than 1 pivot, so can't have $\ge n$ pivots

The case of full column rank → $r=n$

↳ So this means all the columns have pivots

↳ $r=n$ → no free variables

↳ $N(A)=\{\text{zero vector}\}$, null space only has zero vector

↳ Solution to $Ax=b$ → $x=x_p$ (unique solution if it exists)

↳ 0 or 1 solution

ex. $A=\begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 6 & 1 \\ 5 & 1 \end{bmatrix}$ rank=2 $\Rightarrow R=\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

$A=\begin{bmatrix} 1 & 2 & 6 & 5 \\ 5 & 1 & 1 & 1 \end{bmatrix}$ rank=2

$R=\begin{bmatrix} 1 & 0 & - & - \\ 0 & 1 & - & - \end{bmatrix}$

The case of full row rank ($r=m$)

↳ Can solve $Ax=b$ for every b (exists)

↳ every row has pivot

↳ $n-r$ free variables ($n-m$)

The case of $r=m=n$

↳ square matrix, full rank

$A=\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ → invertible ($r=m=n$)

$\boxed{R=I}$ $\boxed{N(A)=0}$ 1 solution

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Lecture #9 - Independence, Basis, and Dimension 12/18/25

Linear Independence - spanning a space - basis and dimension

Suppose A is m by n with $m<n$. (More unknowns than equations)
Then there are nonzero solutions to $Ax=0$
Reason: There will be free variables

* Independence : vectors $x_1, x_2, \ldots, x_n$ are independent if no combinations give zero vector (except the zero combination) since all $c_i = 0$

$$c_1x_1 + c_2x_2 + \ldots + c_nx_n \neq 0$$

Repeated Definition: When $v_1, \ldots, v_n$ are columns of A

(rank=n) ↳ They are independent if nullspace of A is $\left\{ \begin{matrix} \text{zero} \\ \text{vector} \end{matrix} \right\}$ (only 0 vector) → no free variables
(rank<n) ↳ They are dependent if $Ac=0$ for some nonzero c.
(something else in N(A))
↳ yes free variables

* Span : What does it mean for vectors to span a space?

(Vectors $v_1, \ldots, v_l$ span a space) means the space consists of all combinations of those vectors

* Basis for a vector space is a sequence of vectors $(v_1, v_2, \ldots, v_l)$ with 2 properties : 1) They are independent
2) They span the space

↓ Just right

Basis Example : Space is R^3

One basis is $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$ ← standard basis

another basis : $\underbrace{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}2\\2\\5\end{bmatrix}}$

are they independent? Yes

Do they span R^3? NO

we need another

$\underbrace{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}2\\2\\5\end{bmatrix}, \begin{bmatrix}3\\3\\8\end{bmatrix}}_{\text{error}}$

Not invertible

2 identical rows

dependent columns

The test for basis → put them in the columns of matrix and do elimination etc. to see if there exists free variables

For R^n, if we have n vectors that give <u>basis</u>, then the $n \times n$ matrix with those columns is invertible

There exists many many # of basis, but they always have the same # of vectors (n vectors in space R^n)

↑ the number is the dimension of the space

Let's say space is $C(A)$

$\begin{bmatrix}1&2&3&1\\1&1&2&1\\1&2&3&1\end{bmatrix}$ ⇒ Do they span column space of the matrix? Yes

Are they independent? No → not a basis

$\begin{bmatrix}1&2&3&1\\0&-1&-1&0\\0&0&0&0\end{bmatrix}$ $N(A) = \overbrace{\begin{bmatrix}-1\\-1\\1\\0\end{bmatrix}\begin{bmatrix}-1\\0\\0\\1\end{bmatrix}}^{\text{the 2 special solutions}}$ → basis of $N(A)$

↳ <u>Dimension of $N(A)$</u>? = # of free variables $= n - r$

<u>What is the basis?</u> → $2 = \text{rank}(A) = \#$ of pivot columns = dimension of $C(A)$ subspace

→ matrix

→ subspace

Columns 1 & 2 (pivot cols)

can't do 3 or 4

If you know the Dim $C(A) = 2$, and you have 2 independent vectors, they have to span the space

Lector #10 - The Four Fundamental Subspaces 12/19/25

The Four Spaces (subspaces)

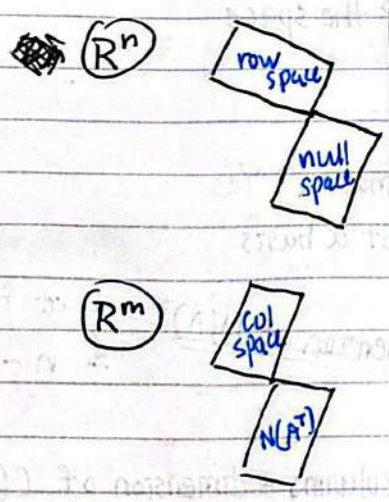
Column Space - $\boxed{C(A)}$
Nullspace - $\boxed{N(A)}$
rowspace - all combinations of rows $\nearrow \boxed{C(A^T)}$
↳ all combinations of cols of A^T
we like working with columns so to get from rows → cols we transpose the matrix
Nullspace of A^T - $\boxed{N(A^T)}$ (left nullspace of A)

$\left\{ \begin{array}{l} C(A) \\ N(A) \\ C(A^T) \\ N(A^T) \end{array} \right.$

Let's say A is $m \times n$, then:

$C(A)$ is in R^m → every col has m components
$N(A)$ is in R^n → nullspace solutions vector has n components
$C(A^T)$ is in R^n → transposed = every col has n components
$N(A^T)$ is in R^m → transposed = solution vector has m components

Picture of 4 subspaces



For each one of these spaces, how would we construct a basis? (systematically) dimension?

	Dimension	Basis
$C(A)$	→ dim $C(A)$ = rank r *	→ Pivot Columns
$C(A^T)$	→ dim $C(A^T)$ = rank r	→ The non-zero rows
$N(A)$	→ dim $N(A)$ = $n-r$	→ Special Solutions
$N(A^T)$	→ dim $N(A^T)$ = $m-r$	→ The bottom rows

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$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \overset{I}{\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}} & \overset{F}{\begin{matrix} 1 & 1 \\ 1 & 0 \end{matrix}} \\ \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} \end{bmatrix} = R \qquad C(R) \neq C(A)$$

different column spaces but same row space?

So the basis for row space (A or R) is the first r rows for R (BUT NOT A)

The row basis is sitting in R in the cleanest form

4th space : $N(A^T)$

$$\boxed{A^T y = 0} \rightarrow \underset{\text{matrix}}{\begin{bmatrix} \quad \end{bmatrix}} \underset{\text{col vector}}{\begin{bmatrix} \quad \end{bmatrix}} = 0$$

$y^T A^{TT} = 0^T$

$\hookrightarrow y^T A = 0^T \rightarrow \underset{\text{row vector}}{\begin{bmatrix} y^T \end{bmatrix}} \underset{\text{matrix}}{\begin{bmatrix} A \end{bmatrix}} = \begin{bmatrix} 0 \end{bmatrix}$

Makes it into row vector but we generally want to stay with cols,

$\boxed{A^T y = 0}$

So stick with this

Idea of taking Matrix A (m by n) and tacking on I matrix (m by m) to get [RE]

$\text{rref}\,[A_{m\times n}\ I_{m\times m}] \rightarrow [R_{m\times n}\ E_{m\times m}]$ $\left\{ \text{In Ch2, R was I, then E was } A^{-1} \right\}$

$\uparrow E$

$\hookrightarrow EA = R$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = E?$$

Check by multiplying this to A

Looking at left nullspace = looking for combination that gives zero row

New Vector Space (M)

↳ All 3×3 matrices (every 3×3 matrix is a vector)

$A+B$, CA → not interested in AB for now for a vector space

Subspaces?

↳ subspaces of M (Matrix Space)

↓

- All upper Δ matrices
- All symmetric matrices
- Diagonal matrices → Dim. of D subspace is 3 → $\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix}}_{\text{spans subspace D}}$

Lecture #11 - Matrix Spaces ; Rank 1 ; Small World Graphs 12/21/25

M = 3 by 3 matrices ← Dim of M = 9

↳ Basis for M Standard Basis : $\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&1&0\\0&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&1\\0&0&0\\0&0&0\end{bmatrix} \ldots \begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}$

<u>Subspaces of M</u>

S Symmetric 3×3 → Dimension = 6

U Upper Δ 3×3 → Dimension = 6

$S \cap U$ = symmetric and upper triangular

= diagonal 3×3's $\dim(S \cap U) = 3$

$S \cup U \Rightarrow$ doesn't work

↳ to get to a bigger space → S+U = any element of S + any element of U

$\dim(S+U) = 9$ = all 3×3 matrices

the dimension of S + the dimension of U = the dimension of their intersection + sum

$$D(S) + D(U) = D(S+U) + D(S \cap U)$$

$\frac{d^2y}{dx^2} + y = 0$

$y = \underbrace{\cos x, \sin x}_{\text{basis}}$, complete solution : $y = C_1 \cos x + C_2 \sin x$

Rank 1 Matrix

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 8 & 5 \end{bmatrix} \rightarrow$$ 2nd row = multiple of 1st row

2×3

The basis for this matrix = 1 4 5 (1st row)

$(r=1)$

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 8 & 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$$

$2\times 3 \qquad 2\times 1 \qquad 1\times 3$

Rank 1 matrix → has some form of $A = uv^T$

↳ they are like the building block for all matrices

→ 5x17 matrix with rank of 4? Will need 4 rank 1 matrices to break it down

M = all 5x17 matrices

(subset of rank 4 matrices) → is this a subspace? Not a subspace

(subset of rank 1 matrices) → is this a subspace? Not a subspace

In R^4 $\quad v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$ $\quad S$ = all v in R^4 with $v_1 + v_2 + v_3 + v_4 = 0$

= nullspace of $A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$ ← rank = 1 = r

$n = 4$

$\dim N(A) = n - r$

$= 3$

Four Fundamental Subspaces of $A = [1, 1, 1, 1]$

Basis for S

$$\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$ → 3 vectors that are the special (natural) to list

$C(A) = R^1$

$N(A^T) = \{0\}$

↓

0-dimension which means no basis

$3 + 1 = 4 = n$

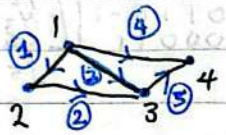
$1 + 0 = 1 = M$

Lecture #12 - Graphs, Networks, Incidence Matrices 12/22/25

Graph = { nodes, edges }

ex.

5 nodes, 6 edges → a 5×6 matrix will tell us everything about that graph



n = 4 nodes

m = 5 edges

→ **Incidence Matrix**

Current flow

node 1 2 3 4

$$A = \begin{array}{c} \\ \\ \\ \\ \\ \end{array}\begin{array}{c} \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \end{array} \begin{array}{l} \text{edge} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array}$$

2, 3 } loop

row 1 + 2 = 3

0 = edge

1 = node

Every row corresponds to an edge

Every col corresponds to a node

What's in the nullspace?

Are the columns independent?

$Ax = 0$

vector $x = x_1, x_2, x_3, x_4$

↳ potentials at nodes

↓

if multiply by A → $x_2 - x_1$, etc

↳ difference in potential across edges

$$Ax = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \\ x_4 - x_1 \\ x_4 - x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

↓

rank = 3

$x = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ → in nullspace & basis for N(A)

set 1 variable (last) = 0

↳ 3 independent cols

whole nullspace $x = c\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$; dim N(A) = 1

What does this mean physically?

Grounding a node is to get it out (setting the potential to 0)

Cont.

cont →

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

TAKEAWAYS

$\dim N(A^T) = m - r = \#$ of loops
$= \#$ of edges $-$ ($\#$ of nodes -1)

$\#$ nodes $-$ $\#$ edges $+$ $\#$ loops $= 1$
Euler's Formula

Most Fundamental Eqn of Applied Mathematics

$A^T y = 0$

↳ finding $N(A^T)$ $\qquad \dim N(A^T) = m - r = 5 - 3 = 2$

$$A^T = n \times m = 4 \times 5 = \begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \qquad A^T y = 0 \rightarrow \begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Why are we interested in this equation?

from prev. page) $\quad x = x_1, x_2, x_3, x_4$ (potentials at nodes)

↳ mult. by A → $e = Ax$

↳ $x_2 - x_1$, etc. (potential differences)

OHM'S LAW ↓ Matrix C → $y = Ce$

currents y_1, y_2, y_3, y_4, y_5 on edges

Kirchoff's Current Law ↓ $A^T y = 0$ → $A^T y = F$

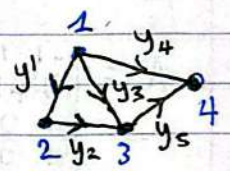
$\boxed{A^T C A x = f}$

$A^T y = 0$

$$\begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

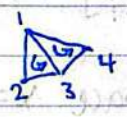
the currents leaving node 1

→ $-y_1 - y_3 - y_4 = 0$
→ $y_1 - y_2 = 0$
→ $y_2 + y_3 - y_5 = 0$
→ $y_4 + y_5 = 0$



Basis of $N(A^T)$

Dim = 2 so 2 vectors ↳ $\begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$



↳ we walk in loop of 1234 and 134
these are independent ⇒ basis

↳ the loop of 1234 ▱ is not independent but rather a (comb.) sum of 123 & 134

$C(A^T)$ → rowspace of A^T

cols 1, 2, 4 are the pivot cols
↓
the cols that DONT have a loop (independence)

$\dim N(A^T) = m - r$
↳ $\#$ of loops $= \#$ of edges $- r$ (rank $= n-1$)

A graph without a loop = TREE

Lecture #13 - ~~Quiz~~ Exam 1 Review 12/23/25

1) Suppose u, v, w (non-zero vectors) in R^7, they span a subspace of R^7. What are the possible dimensions?

<u>1, 2, or 3</u>

2) 5×3 matrix u, $r=3$, what's the nullspace?

$N(u) = \{\begin{bmatrix}0\\0\\0\end{bmatrix}\}$ → reasoning: u has 3 columns (5×3)

rank $=3 \Rightarrow$ 3 pivot columns

all cols are independent → $\begin{bmatrix}0\\0\\0\end{bmatrix}$ vec

↳ no comb. of cols $=0$ ($ax=0$)

3) 10×3 matrix $B = \begin{bmatrix}u\\2u\end{bmatrix}$, what's the rank & echelon form?

reduced

echelon form → $\begin{bmatrix}u\\0\end{bmatrix}$ rank=3

10×6 matrix $C = \begin{bmatrix}u & u\\u & 0\end{bmatrix} \to \begin{bmatrix}u & 0\\0 & -u\end{bmatrix} \to \begin{bmatrix}u & 0\\0 & +u\end{bmatrix}$ rank=6

$\dim N(C^T) = m-r = 10-6 = 4$

4) $Ax = \begin{bmatrix}2\\4\\2\end{bmatrix}$, $x = \begin{bmatrix}2\\0\\0\end{bmatrix} + c\begin{bmatrix}1\\1\\0\end{bmatrix} + d\begin{bmatrix}0\\0\\1\end{bmatrix}$, given eqn of Ax and its complete solution, what is the dimension of the row space?

3 components 3 components

$A = 3\times3$ matrix $\dim N(A) = 2$ (2 independent vectors $c\begin{bmatrix}1\\1\\0\end{bmatrix} + d\begin{bmatrix}0\\0\\1\end{bmatrix}$

$A = \begin{bmatrix}1 & -1 & 0\\2 & -2 & 0\\1 & -1 & 0\end{bmatrix}$ rank $=1$ → <u>$\dim C(A^T) = 1$</u>

3) $Ax=b$ can be solved if...

↳ b has the form $b = c\begin{bmatrix}1\\2\\1\end{bmatrix}$, multiple of $\begin{bmatrix}1\\2\\1\end{bmatrix}$

don't forget the other cases $r=m$, $r=n$ (full rank, etc.)

5) If $N(A) = \{0\}$, what about $N(A^T)$, given square matrix

$N(A^T)$ is also zero vector

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6) 5×5 matrix as a vector, look at the invertible matrices, do they form subspace?

No, because the zero vector is not included

+ adding 2 inv matrices ≠ invertible matrice

7) True <u>False</u> $B^2=0 \Rightarrow B=0$, if $B^2=0$, is $B=0$?

False → ex. $B=\begin{bmatrix}0&1\\0&0\end{bmatrix} \Rightarrow B^2=\begin{bmatrix}0&0\\0&0\end{bmatrix}$

8) n×n matrix, independent cols, is $Ax=b$ always solveable?

↓

rank = n (fullsize) → Yes

9) $B=\underbrace{\begin{bmatrix}1&1&0\\0&1&0\\1&0&1\end{bmatrix}}_{3\times3}\underbrace{\begin{bmatrix}1&0&-1&2\\0&1&1&-1\\0&0&0&0\end{bmatrix}}_{3\times4}$ given B, give a basis of N(B)

↳ matrix would be 3×4 → basis for N(B) in R^4 = ?

dim = ?

↳ invertible

↳ so $N(CD)=N(D)$ if C invertible

bc if invertible, $N(C)=\{0\}$

so just find N(D)

basis for $N(B)=\begin{bmatrix}-1\\1\\0\end{bmatrix},\begin{bmatrix}-2\\1\\0\\1\end{bmatrix}$ ↲ special solutions

Now complete solution to $Bx=\begin{bmatrix}1\\0\\1\end{bmatrix}$

$$x_p+x_n=\begin{bmatrix}1\\0\\0\\0\end{bmatrix}+c\begin{bmatrix}1\\-1\\1\\0\end{bmatrix}+d\begin{bmatrix}-2\\1\\0\\1\end{bmatrix}$$

How to get x_p?

first col of B

↳ $\underbrace{\begin{bmatrix}1&1&0\\0&1&0\\1&0&1\end{bmatrix}}_{C}\underbrace{\begin{bmatrix}1\\0\\0\end{bmatrix}}_{1^{st}\text{ col }D}=\overset{RHS}{\begin{bmatrix}1\\0\\1\end{bmatrix}}$

first col B = vector B

↳ only need 1 col of first col, 0 of others

$x_p=\begin{bmatrix}1\\0\\0\end{bmatrix}$

(this is not the only particular x_p but we take the most convenient)

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